Practice Set 22 SAQ Sheet – Short Answer Practice

1. Q3a. Noonan syndrome
2. Q3b. Neurogenic bladder in myelodysplasia
3. Q3c. Eponymous techniques of VVF repair
4. Q3d. Endometriosis in urology
5. Q3e. Hypospadias repairs – single-stage versus staged techniques
6. Q3f. Instestinal segments in urology (repeat of 2022-Summer Paper 1, Q3a)

Short-answer practice sheet compiled from handwritten study lists.

Q3a. Noonan syndrome

Type: SAQ (10 marks)

Definition and genetics

- Noonan syndrome is an **autosomal dominant RASopathy** (disorder of the RAS–MAPK [mitogen-activated protein kinase] signalling pathway) with variable expression; many cases are de novo. Incidence about 1 in 1,000–2,500.
- **Karyotype is normal** (46,XY or 46,XX). The phenotype resembles Turner syndrome, hence the old name "male Turner".
- **Mechanism:** gain-of-function variants increase RAS–MAPK signalling, which disturbs growth and organ development.

Gene	Approximate share	Note
PTPN11 (encodes SHP2 phosphatase)	About 50%	Pulmonary stenosis common
SOS1	About 10%	Milder growth and learning effects
RAF1	About 5–10%	Hypertrophic cardiomyopathy
RIT1, KRAS, others	Smaller	KRAS: more learning difficulty

- Related RASopathies: Costello, cardiofaciocutaneous, NF-1 (neurofibromatosis type 1).

Clinical features

- Short stature; webbed neck; hypertelorism, ptosis, low-set posteriorly rotated ears; pectus deformity.
- **Heart:** pulmonary valve stenosis (commonest), HCM (hypertrophic cardiomyopathy), septal defects.
- **Bleeding tendency:** factor XI deficiency, platelet dysfunction.
- Lymphatic dysplasia: genital (penoscrotal) lymphoedema; mild learning difficulty.

Urological aspects

- **Cryptorchidism:** the main urological feature, in most affected boys (often bilateral).
- **Testicular dysfunction:** primary Sertoli cell dysfunction (low inhibin B, high FSH [follicle-stimulating hormone]) with poor spermatogenesis, even after orchidopexy. Leydig function is usually adequate; puberty is often delayed. Female fertility is usually normal.
- **Renal anomalies (about 10%):** mostly mild renal pelvic dilatation; also duplex system, solitary kidney, hypoplasia.
- Small excess of childhood cancers (JMML [juvenile myelomonocytic leukaemia], embryonal rhabdomyosarcoma).

Evaluation

- Clinical diagnosis (van der Burgt criteria), confirmed by gene panel; karyotype excludes Turner in girls.
- Before surgery: **echocardiography, coagulation screen and factor XI level**, anaesthetic review.
- Renal and bladder ultrasound at diagnosis.

Management

- **Orchidopexy by 18 months** (EAU [European Association of Urology] and AUA [American Urological Association]); laparoscopic for non-palpable testes.
- Follow-up: pubertal hormones, semen analysis, fertility counselling.

- **Also know:** growth hormone for short stature; multidisciplinary cardiac and haematology care.

Diagram to draw: RAS–MAPK cascade with SHP2 (PTPN11) acting upstream of RAS.

> **Key points to remember:**
> - Autosomal dominant, normal karyotype; PTPN11 in about half.
> - Cryptorchidism plus Sertoli dysfunction: male subfertility.
> - Check heart and clotting (factor XI) before orchidopexy.

Q3b. Neurogenic bladder in myelodysplasia

Type: SAQ (10 marks) (prenatal and neonatal assessment also in 2016-Summer Paper 3, Q3e)

Introduction

- Myelodysplasia (NTD [neural tube defect], mainly **myelomeningocele**) is the commonest cause of neurogenic LUT (lower urinary tract) dysfunction in children. Lesions: lumbosacral 47%, lumbar 26%, sacral 20%.
- **Bony or motor level does not predict bladder function**; the lesion can progress (re-tethering), so urodynamic testing is essential.
- About 10% of deaths in SB (spina bifida) are urological. Folic acid 400 µg daily before conception prevents 50–70% of NTDs.

Patterns

- Detrusor: overactive (NDO, neurogenic detrusor overactivity) or underactive. Outlet: competent, incompetent or dyssynergic.
- Newborns: most have detrusor contractions; over a third are acontractile. Needle EMG (electromyography): intact sacral arc about 40%, partial denervation 25%, complete 35%.

Evaluation (EAU [European Association of Urology] 2022)

- Renal and bladder ultrasound after birth, then serially; **UDS (urodynamics) within 3 months**; baseline DMSA (dimercaptosuccinic acid); creatinine (cystatin C better with low muscle mass); VCUG (voiding cystourethrogram) or video-UDS for VUR (vesicoureteric reflux), which is dilating in 7%.

Risk (McGuire 1981)	Finding
Hostile	DLPP (detrusor leak point pressure) or end-filling pressure ≥40 cm H_2O
Intermediate	NDO, poor compliance, 25–39 cm H_2O
Safe	Below 25 cm H_2O

Management (kidneys first, then continence)

- **Early CIC (clean intermittent catheterisation; Lapides 1971) plus oxybutynin** for hostile bladders. Hydronephrosis 0–15% versus 18–80% expectant; ESRD (end-stage renal disease) 0–1.6% versus about 30%.
- Others: solifenacin, mirabegron; **botulinum toxin A** about 10 U/kg (maximum 360 U), continence 65–87%.
- Surgery: vesicostomy (bridge); **augmentation cystoplasty**; Mitrofanoff channel; bladder-neck sling, reconstruction or artificial sphincter.
- VUR: CIC and antimuscarinic first; avoid Credé; reimplant during augmentation if grade ≥III.
- **Also know:** latex allergy; bowel programme; sexual dysfunction (about 75% of men have ED [erectile dysfunction]; sildenafil helps); precocious puberty.

Diagram to draw: UDS trace with DLPP above 40 cm H_2O.

> **Key points to remember:**
> - Level does not predict bladder: UDS within 3 months.
> - DLPP ≥40 cm H_2O = hostile (McGuire 1981).
> - Early CIC plus antimuscarinic protects kidneys.

Q3c. Eponymous techniques of VVF repair

Type: SAQ (10 marks) (management of VVF also in Practice Set 3 Paper 3, Q2)

Introduction

- VVF (vesicovaginal fistula): an epithelialised bladder–vagina tract. **Principles:** exclude ureteric involvement and cancer; tension-free, multilayer, non-overlapping, watertight closure; interposition when needed; drainage ≥10–14 days. **The first repair has the best chance.**

Transvaginal techniques

- **Sims:** first successful systematic vaginal repair (silver wire).
- **Flap-splitting (Raz)**:
  - Stent ureters if close; Foley through the tract for traction.
  - J or U incision with a 1–3 mm cuff; raise anterior and posterior vaginal flaps.
  - Layer 1: bladder (3-0 or 4-0), with the tract edge. Layer 2: perpendicular perivesical layer. Fill test 200–300 mL.
  - Optional Martius or peritoneal flap; advance the opposite vaginal flap ≥2 cm beyond. Closure ≥90%.
- **Latzko (1942), high partial colpocleisis**:
  - Denude vaginal epithelium 1–2 cm around the tract; the bladder is **not opened or dissected**.
  - Close vaginal walls over the tract in layers.
  - ≥90% success in non-irradiated post-hysterectomy apical VVF; favoured near ureters or after radiation. Drawback: **vaginal shortening and dyspareunia**.
- **Also know:** vaginal cuff excision (all apical scar excised as an inverted cone).

Abdominal technique

- **O'Conor and Sokol (1951)**, suprapubic extraperitoneal:
  - Midline cystotomy at the dome, **bivalved down** to the fistula; ureteric catheters.
  - Excise the tract; separate bladder from vagina 2–3 cm beyond.
  - Close vagina; omentum or peritoneum between; close bladder in layers; suprapubic catheter.
- Modifications: limited-cystotomy transvesical; extravesical. Laparoscopic (1994) and robotic (2005) versions give equal closure with less morbidity.

Interposition flaps

Flap	Blood supply	Use
Martius (1928), labial fibrofatty pad	Internal pudendal (inferior), external pudendal (superior)	Mid–distal VVF; does not reach the apex
Peritoneal	Local	High apical VVF
Omental J-flap	Gastroepiploic	Abdominal, radiation
Gracilis, Singapore	Myocutaneous	Refractory cases

- RCT (randomised controlled trial): Martius adds nothing in simple VVF.

Diagram to draw: Latzko denuded area versus flap-splitting layers.

Key points to remember:

- Latzko: colpocleisis, bladder not opened, vagina shortened.
- O'Conor: bivalve bladder to the fistula, interpose omentum.
- Martius for distal, omentum or peritoneum for apical VVF.

Q3d. Endometriosis in urology

Type: SAQ (10 marks)

Definition and epidemiology

- Endometriosis is endometrial glands and stroma outside the uterus. Urinary tract involvement occurs in about 1–5% of women with endometriosis, mostly with deep disease.
- Sites: **bladder** (commonest, about 80%) > ureter (about 15%) > kidney, urethra (rare).

Pathogenesis

- Retrograde menstruation (Sampson), coelomic metaplasia, lymphovascular spread, and **iatrogenic implantation** at pelvic surgery (e.g. caesarean section).
- Bladder lesions start on the **serosa** (vesicouterine pouch) and invade the detrusor.
- Ureteric disease: **extrinsic** (periureteric fibrosis, commoner) or intrinsic (muscularis or mucosa); usually distal, left more often.

Clinical features

- **Bladder:** cyclical haematuria or frequency, urgency and dysuria with menses; suprapubic pain. Differential: BPS (bladder pain syndrome), vesicouterine fistula (menouria).
- **Ureter:** often **silent**; hydronephrosis and loss of renal function may go unnoticed.

Evaluation

- Transvaginal and transabdominal ultrasound gauge size and depth; **renal ultrasound in every woman with deep endometriosis**.
- **MRI (magnetic resonance imaging):** extent; high T1 signal from blood products.
- **Cystoscopy:** limited value (disease is extravesical); may show a bluish submucosal lesion. Biopsy excludes carcinoma.
- CTU (computed tomography urography) and MAG3 (mercaptoacetyltriglycine) renogram for obstruction and split function.
- Staging: rASRM (revised American Society for Reproductive Medicine) I–IV; #Enzian records bladder and ureter.

Management (with gynaecologist)

- **Medical:** progestins (dienogest), combined pill, GnRH (gonadotrophin-releasing hormone) analogues. Symptomatic relief; not for obstructive ureteric disease.
- **Bladder: partial cystectomy** (full-thickness excision with margin), laparoscopic or robotic; stent ureters if near. TUR (transurethral resection) is incomplete, with recurrence and perforation risk.
- **Ureter:**
  - Extrinsic, mild: ureterolysis.
  - Intrinsic or stricture: segmental resection with ureteroureterostomy, or **reimplantation with psoas hitch or Boari flap**.
  - Non-functioning kidney: nephrectomy.
- Treat all pelvic disease at the same operation to lower recurrence.
- **Also know:** risk factor for VVF (vesicovaginal fistula) and ureteric injury at hysterectomy; rare malignant change (clear-cell, endometrioid).

Diagram to draw: endometriotic nodule in the vesicouterine pouch invading the detrusor.

Key points to remember:

- Bladder commonest; cyclical haematuria.
- Ureteric disease silent: screen kidneys.
- Partial cystectomy, not TUR.

Q3e. Hypospadias repairs – single-stage versus staged techniques

Type: SAQ (10 marks) (present concepts also in Practice Set 6 Paper 4, Q1)

Introduction

- Choice depends on **curvature after degloving**, plate quality and glans size, not on meatal position alone.
- Algorithm: deglove, artificial erection. Straight or <30°: dorsal plication, single-stage. >30°: divide plate, ventral corporotomy ± graft, then staged.

Single-stage techniques

Group	Examples	Indication
Advancement	**MAGPI** (meatal advancement and glanuloplasty, Duckett 1981); MIV (M inverted V)	Glanular meatus, no curvature
Tubularisation	**TIP** (tubularised incised plate, Snodgrass 1994); Thiersch–Duplay; GAP (glans approximation procedure)	Distal–midshaft; TIP in >90% of distal
Flap	Mathieu (flip-flap); **OIF** (onlay island flap)	Thin plate
Proximal one-stage	TPIF (transverse preputial island flap, Duckett tube); proximal OIF	Curvature corrected; good tissue
Graft	Dorsal buccal inlay	Narrow plate

- Pros: one anaesthetic. Proximal TIP is abandoned: devascularised plate, stricture, recurrent curvature.

Staged techniques

- Now the approach of choice for most proximal hypospadias. **Rationale:** a urethroplasty over a fresh corporal graft fails; stage 1 builds a supple, vascular bed.

Type	Stage 1	Stage 2 (≥6 months)
Bracka / STAG (staged tubularised autograft)	Correct curvature, divide plate, quilt inner preputial or buccal graft	Tubularise, glansplasty, tunica vaginalis cover
Byars flaps	Split dorsal prepuce, transpose ventrally	Tubularise
Distal-first (Ulaanbaatar)	Curvature and distal urethra	Join proximal urethra

- Repeat artificial erection before stage 2.

Comparison of outcomes

Measure	Single-stage	Staged
Distal success	83–95%	–
Proximal complications	About 60% (long TIP, mixed)	STAG about 23%; Byars 43–81%
Recurrent curvature	Higher after plication	0–10%
Penile length, perception	Lower	Better
Extra procedures	Fewer planned	25–75% need more

- Fistula is the commonest complication. Long, rigid neourethra without spongiosum raises resistance (Poiseuille's law), causing fistula and diverticulum.

Diagram to draw: STAG stage 1 graft and stage 2 tubularisation.

Key points to remember:

- Curvature >30° after degloving: stage.
- TIP for distal; STAG has lowest proximal complications.
- Stage 2 after ≥6 months.

Q3f. Instestinal segments in urology

Type: SAQ (10 marks) (also answered in 2022-Summer Paper 1, Q3a)

Assumed meaning: intestinal segments used in urinary diversion and bladder reconstruction.

Introduction

- Bowel is used for conduits, continent pouches, neobladders, augmentation and ureteric replacement. Choice depends on renal function, irradiation, previous surgery, mesenteric length and the reconstruction needed.

Segments compared

Segment	Electrolyte disturbance	Advantage	Drawback
Stomach	Hypochloraemic, hypokalaemic metabolic alkalosis	Secretes acid, less mucus; for renal failure, short gut	Haematuria–dysuria syndrome, ulcers
Jejunum	Hyponatraemic, hypochloraemic, **hyperkalaemic** acidosis	Outside radiation field	Severe salt loss; avoid
Ileum	**Hyperchloraemic metabolic acidosis**	Mobile, constant supply; default conduit	B12 and bile-salt loss; obstruction ~10%
Colon	Hyperchloraemic metabolic acidosis	Easy antireflux tunnel; transverse colon spared by pelvic radiation	Diarrhoea if ileocaecal valve used

- **Mechanism (ileum, colon):** NH_4^+ replaces Na^+ on the Na^+/H^+ exchanger, coupled with Cl^-/HCO_3^- exchange, so ammonium chloride is absorbed. Treat with sodium bicarbonate or citrate (potassium citrate if hypokalaemic); chlorpromazine or nicotinic acid limit chloride transport.

Other complications

- **Hypokalaemia** (worse with colon); correct potassium before acidosis.
- **Altered sensorium:** hyperammonaemia (drain diversion, lactulose), hypomagnesaemia, drug reabsorption (phenytoin, methotrexate).
- **Osteomalacia:** acidosis, vitamin D resistance, calcium loss; growth retardation in children.
- **Infection, stones** (ileal conduit 10–12%, continent caecal reservoir ~20%), short-gut effects.
- **Cancer:** ureterosigmoidostomy 6–29% after 10–20 years; endoscopy from 5 years.

Renal function needed for continent diversion

- Creatinine <2 mg/dL; GFR (glomerular filtration rate) >35 mL/min; urine pH ≤5.8 after ammonium chloride load; osmolality ≥600 mOsm/kg.

Technical principles

- Clear ≤8 cm of mesentery from a segment end; avoid watershed areas (Griffiths, Sudeck points).
- **Also know:** ERAS (enhanced recovery after surgery) with no routine mechanical bowel preparation; Wallace (stricture ~3%) and Bricker anastomoses; nipple stoma.

Diagram to draw: ileal loop with mesenteric arcade and Wallace anastomosis.

Key points to remember:

- Ileum and colon: hyperchloraemic acidosis (NH_4Cl absorption).
- Jejunum: hyperkalaemia; stomach: hypochloraemic alkalosis.
- Continent diversion: GFR >35, pH ≤5.8, osmolality ≥600.

Rapid Revision Sheet – Practice Set 22 SAQ Sheet

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

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- Cryptorchidism plus Sertoli dysfunction: male subfertility.
- Check heart and clotting (factor XI) before orchidopexy.

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- Level does not predict bladder: UDS within 3 months.
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